

Online Supplementary Materials

Modelling Classroom Co-Regulation from LMS Interaction Logs for Real-Time Classroom Orchestration: A Psychometrically Validated Trivariate State Framework

S1. Annotation Handbook

This supplementary section provides the full annotation handbook referenced in Section 4.4 of the main article, including the four protocol rule cards in tabular form, the calibration procedure, illustrative cases constructed to demonstrate protocol application, and rating distributions on the $N = 400$ annotated sample. The protocols below are reproduced in the form used during calibration; final ratings reflected structured expert judgement guided by these references rather than deterministic procedural application, as discussed in Section 4.4 of the main article.

S1.1. Annotation rule cards

Tables 1–4 collect the reference thresholds and decision aids used during annotation. The four protocols share a common form: each provides indicative anchors that orient rater judgement, but raters were instructed to integrate evidence holistically rather than to apply the rules deterministically.

Table 1: Alignment protocol rule card. The four-element evidence structure parallels the four channels of the alignment proxy (Section 4.3 of the main article). Annotators integrated evidence across all four elements, with the cascade-bundled event-type concentration (C_e) providing the initial anchor.

Evidence element	Reference threshold
(1) Event-type concentration C_e	$\geq 0.90 \rightarrow 5$; $0.75\text{--}0.89 \rightarrow 4$; $0.50\text{--}0.74 \rightarrow 3$; $0.30\text{--}0.49 \rightarrow 2$; $< 0.30 \rightarrow 1$
(2) Component-level corroboration C_c	Agreement within ± 0.10 of C_e : preserve rating. Divergence > 0.20 : anchor to higher-concentration channel.
(3) Substantive vs. generic activity	Substantive (learning task): preserve rating. Generic (navigation, idle): weaken by 1 level.
(4) Temporal sustainment	Both adjacent windows match dominant activity: strengthen by 1. One matches: preserve. Neither matches: weaken by 1.
<i>Final rating bounded to $\{1, 2, 3, 4, 5\}$.</i>	

Table 2: Dispersion protocol rule card. Annotators counted distinct activity-type families after cascade bundling and mapped the count to the 1–5 ordinal scale.

Distinct activity types K	Rating
1	1
2	2
3	3
4	4
≥ 5	5

S1.2. Calibration procedure

A half-day calibration session preceded independent annotation. The session was structured in five blocks: (1) a 45-minute walk-through of construct definitions, the rule cards above, and worked examples; (2) a 90-minute first-pass independent annotation of 25 calibration windows drawn from the corpus

Table 3: Stability protocol rule card. Three continuity checks orient the rating; mild-versus-sharp change distinguishes the two lower categories. End-of-session participation decay with preserved task focus was rated as a stable tapering process rather than as a regulatory rupture.

Continuity check	Anchor
(1) Prior-window match	Dominant activity and active-user count match prior window.
(2) Next-window match	Dominant activity and active-user count match next window.
(3) Temporal continuity	Timestamps continuous between retained windows.
Number of affirmative checks	Rating
3 affirmatives	5
2 affirmatives	4
1 affirmative	3
0 affirmatives, mild between-window changes	2
0 affirmatives, sharp between-window changes	1

but disjoint from the main 400-window task; (3) a 60-minute facilitated discussion of disagreements, with consensus reasoning recorded for each discussed window; (4) a 60-minute re-annotation of the 25 calibration windows; (5) a qualification check requiring each annotator to show pairwise Pearson $r \geq 0.70$ with at least three of the other three annotators on each dimension. All four annotators met the qualification criterion. The calibration discussion focused particularly on boundary cases: the substantive-versus-generic distinction for the alignment protocol, the count-versus-bundle decision for the dispersion protocol, and the mild-versus-sharp change distinction for the stability protocol.

S1.3. Illustrative cases

The cases below are *illustrative cases constructed to demonstrate protocol application*. They are not drawn from the empirical corpus; they are synthesised to show how the four protocols integrate evidence in characteristic situations. Empirical rating distributions across the $N = 400$ annotated windows are reported separately in Section S1.4 below.

Illustrative case A: high alignment in a problem-solving context.. A 60-second window during a problem-solving session shows 12 active users, with 11 of 12 producing events on a single quiz-attempt activity (after cascade bundling collapses the underlying logging cascade into one event type). The dominant Moodle component is `mod_quiz`; the second-most-frequent event type accounts for 4% of events. Both the prior and next windows share the same dominant activity and active-user count.

Protocol application: Event-type concentration $C_e \approx 0.96$ places the alignment ladder anchor at rating 5. Component-level concentration $C_c \approx 0.96$ corroborates within ± 0.10 , so the rating stands. The dominant activity is substantive (quiz attempts on a problem-solving task), so the rating is preserved. Temporal sustainment: both adjacent windows match, strengthening by 1 — capped at the ceiling. **Alignment rating: 5.** Dispersion count: 1 distinct activity type, mapping to **dispersion rating: 1.** Stability: all three continuity checks affirmative, mapping to **stability rating: 5.** Indicative action: *monitor*.

Illustrative case B: moderate alignment with end-of-session decay.. A 60-second window late in an independent-work session shows 6 active users (down from 11 in the prior window), distributed across 3 activity types after cascade bundling: file views (42%), forum reads (33%), and assignment-page views (25%). The dominant activity (file views) is generic navigation rather than a substantive learning task. The prior window had the same dominant activity but with 11 users; the next window shows 4 users and a different dominant activity (forum reads). Timestamps are continuous.

Protocol application: Event-type concentration $C_e \approx 0.42$ places the alignment ladder anchor at rating 2. Component-level concentration is close to C_e , so the rating stands. The dominant activity is

Table 4: Action protocol rule card. The six instructional action categories and indicative state combinations that guided selection. Annotators applied professional judgement for ambiguous or boundary conditions; the indicative state column is descriptive rather than prescriptive.

Action category	Indicative state combination
<i>monitor</i>	High alignment, low dispersion, high stability: class operating coherently; no intervention needed.
<i>whole_class_prompt</i>	Moderate alignment, moderate dispersion: brief refocusing nudge may help reconverge attention.
<i>clarify_instruction</i>	Low alignment, high dispersion, low stability: signs of widespread misinterpretation of the task.
<i>pacing_adjust</i>	High alignment, moderate dispersion, declining stability: pacing mismatch between class and task.
<i>target_subgroup</i>	High overall alignment, asymmetric participation: a subset of students drifting while majority on task.
<i>pause_and_reset</i>	Low alignment, high dispersion, sharply low stability: regulatory rupture; full reset warranted.

generic (file viewing without substantive engagement), weakening the rating by 1 to rating 1. Temporal sustainment: prior window matches dominant activity, next does not — one match, so the rating stands. **Alignment rating: 1.** Dispersion: 3 distinct activity types, mapping to **dispersion rating: 3.** Stability checks: prior match yes (dominant activity), next match no, temporal continuity yes; 2 affirmatives total. But the participation drop from 11 to 6 to 4 is a sharp change — annotator judgement may downgrade from rating 4 toward rating 3 given the declining engagement, though the end-of-session decay rule (preserved task focus with declining participation rated as stable tapering) provides counter-pressure. **Stability rating: 3 (annotator judgement at boundary).** Indicative action: *clarify_instruction* or *pause_and_reset* depending on session timing.

Illustrative case C: stable navigation context.. A 60-second window in a navigation context shows 8 active users distributed across 5 activity types: home page (28%), course main page (24%), resource list (20%), grade book (16%), profile page (12%). The dominant activity is generic navigation. Prior and next windows show similar distributional patterns with the same five activity types in slightly shuffled proportions.

Protocol application: Event-type concentration $C_e \approx 0.28$ places the alignment ladder anchor at rating 1. Component concentration agrees. Dominant activity is generic, but the rating is already at the floor and cannot be weakened further. Temporal sustainment: both adjacent windows broadly match the distributional pattern (though no single dominant activity matches strictly), so annotator judgement may apply +1 or hold at floor. **Alignment rating: 1.** Dispersion: 5 distinct activity types, mapping to **dispersion rating: 5.** Stability: dominant-activity match is ambiguous because navigation does not have a clear single dominant activity; activity-type continuity is high (same five types present), but the strict prior-match and next-match checks are weak. Annotator judgement integrates the consistent distributional pattern across windows and rates this as **stability rating: 4 (consistent navigation patterning across windows).** Indicative action: *monitor* (low coordination demand) or *whole_class_prompt* (to redirect to substantive activity).

S1.4. Rating distributions on the annotated sample

Table 5 reports the rating distributions across the $N = 400$ annotated windows for each dimension and the action recommendation. These are aggregated across all four raters (i.e., counts shown are out of $400 \times 4 = 1600$ rating events per dimension).

Table 5: Rating distributions across the $N = 400$ annotated sample, aggregated across all four raters. Counts are out of 1,600 rating events per dimension ($400 \text{ windows} \times 4 \text{ raters}$).

Dimension	Rating 1	Rating 2	Rating 3	Rating 4	Rating 5
Alignment	12	45	264	498	781
Dispersion	15	143	826	580	36
Stability	136	439	641	303	81
<i>Action recommendation counts (six categories):</i>					
<i>monitor</i> : 539	<i>whole_class_prompt</i> : 113		<i>clarify_instruction</i> : 58		
<i>pacing_adjust</i> : 336	<i>target_subgroup</i> : 523		<i>pause_and_reset</i> : 31		

The validation analyses reported in Section 5 of the main article use the four-rater consensus mean for each window-dimension pair; the consensus distribution mirrors the per-rater distributions reported above with reduced variance and quarter-integer increments arising from the 4-rater averaging.

S2. Action Agreement Supplementary Materials

This supplementary section provides the full action co-occurrence matrix referenced in Sections 4.5 and 5.4 of the main article. The action protocol assigned one of six instructional action recommendations to each window: *monitor*, *whole_class_prompt*, *clarify_instruction*, *pacing_adjust*, *target_subgroup*, and *pause_and_reset*. The action recommendation was rated independently of the three regulatory dimensions and constitutes a downstream judgement informed by but not deterministically derived from the trivariate state proxies.

S2.1. Overall agreement statistics

Inter-rater agreement on the action recommendation, aggregated across the $N = 400$ annotated windows and four fully-crossed raters, achieved Fleiss’ $\kappa = .686$ across the six action categories (Landis and Koch, 1977). All four raters agreed on the recommended action in 73.0% of windows; at least three of four raters agreed in 89.0% of windows. By Landis-Koch benchmarks (Landis and Koch, 1977), this constitutes substantial agreement on a six-category instructional judgement task.

S2.2. Action co-occurrence matrix

Figure 1 shows the full 6×6 action co-occurrence matrix, computed as the count of rater-pair (window) instances in which one rater chose row category i and another rater chose column category j , summed across all $\binom{4}{2} = 6$ rater pairs and all 400 windows. Row-normalised proportions show within-row agreement and patterns of systematic substitution.

S2.3. Substitution patterns

The matrix reveals systematic substitution patterns rather than diffuse confusion across categories. Two categories show high within-row agreement: *monitor* and *whole_class_prompt* both have row-normalised diagonals above 87%, indicating that when one rater selected either of these actions, the modal choice of the other raters was the same action. These two categories sit at the low-intervention end of the action taxonomy, where the regulatory state evidence is relatively unambiguous.

The principal off-diagonal pattern involves *pacing_adjust*, which is confused with *monitor* in 69% of cases. This boundary admits substantial interpretive variability: distinguishing between a class operating coherently (warranting *monitor*) and a class operating coherently but at a pace mismatched with the

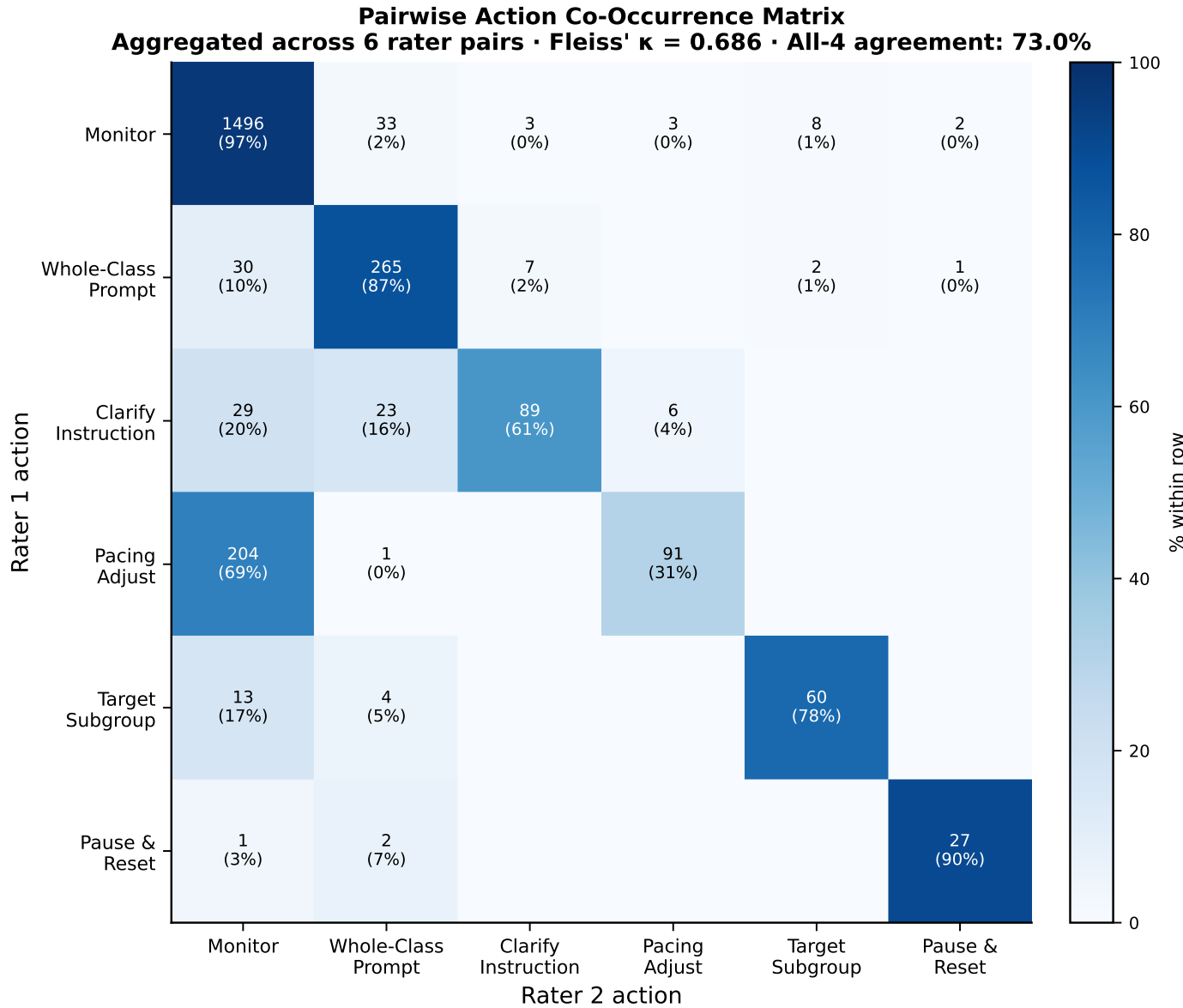


Figure 1: Six-by-six action co-occurrence matrix across the four-rater fully-crossed annotation panel, row-normalised within each row category.

task (warranting *pacing_adjust*) requires a judgement about the intended task timing that is not fully recoverable from the LMS log evidence alone. The action protocol acknowledges this boundary explicitly (Section S1.1, Table 4) and instructs annotators to apply professional judgement.

Other off-diagonal entries are smaller in magnitude and do not exhibit the same systematic pattern. The *clarify_instruction*, *target_subgroup*, and *pause_and_reset* categories show within-row agreement in the 60–75% range with no single dominant confusion partner, suggesting that disagreements on these categories reflect distributed interpretive variability rather than a specific boundary admitting systematic confusion.

S2.4. Implications for action validation

The substantial overall κ combined with the localised *pacing_adjust*–*monitor* confusion has two implications for future validation work. First, the action taxonomy is broadly reproducible at the all-four agreement level (73%), supporting its use as a candidate action vocabulary for downstream decision-support systems built atop the validated state representation. Second, the specific *pacing_adjust*–*monitor* boundary warrants either taxonomic refinement (e.g., introducing an intermediate category for pace-monitoring) or explicit decision-support guidance to teachers about how to distinguish the two situations in practice. The latter is the more conservative path, preserving the six-category structure

validated here while acknowledging the interpretive boundary in deployment-time guidance.

References

Landis, J.R., Koch, G.G., 1977. The measurement of observer agreement for categorical data. *Biometrics* 33, 159–174. doi:[10.2307/2529310](https://doi.org/10.2307/2529310).